Session 8. Model scenario design

In this session, you will work in small groups to develop meaningful model scenarios for the Skive Fjord ecosystem model. These scenarios should reflect the types of questions and management issues that local authorities might face.

Exercise

1. **Brainstorm potential management scenarios**
   In groups of 2-3, identify a range of relevant scenarios that could help inform the coastal waterbody council of Skive Fjord about the fjord's future. You may draw on your background knowledge, course readings, and any other credible external sources.

   Some questions to consider:
   "Climate Change"
   What are the major stressors currently affecting Skive Fjord and Danish coastal water bodies in general? oxygen conditions / temp / salinity spike / storm scenarios
   What types of management interventions are realistic?
   "Salt Spike"
   Are there future pressures that should be explored? Ocean Heat Waves / Ocean Acidification
   What information would coastal managers need to make forward-looking decisions? Oxyg / Chl-a / Mixing
   What processes or components of the ecosystem are most sensitive to change?
   Fish? / Sediment flux / Air Sea Exchange / Phytoplankton

2. **Select and formulate your top three scenarios**
   Choose the three most relevant scenarios from your brainstorm.
   a. Formulate each scenario as a clear hypothesis to be investigated with the 3D hydrodynamic-biogeochemical Skive Fjord model. Include:
      - the specific change(s) to the model configuration (forcing, loads, parameters, climate signal, etc.)
      - the expected model response
      - the temporal and spatial resolution needed in the outputs to test the hypothesis
   b. Justify why each scenario matters from a water-management perspective.

   "Seasonal Nutrient Input"
   "Climate Change Temp"
   "Oxygen Source/Pump"

   Please consider:
   What specific change does your scenario impose (e.g., 20% reduction in nitrogen load, +2°C increase in temperature)?
   What measurable ecosystem response do you expect (e.g., lower summer chlorophyll-a, increased bottom hypoxia)?
   Is the scenario testable with the available model structure and data?
   Is your hypothesis clearly formulated and falsifiable?

"Making the point sources for nutrients seasonal will increase Oxygen condition in the bottom layer"

3. **Present and discuss**
   Share your scenarios with the rest of the class. Be prepared to explain your reasoning and engage in discussion.

4. **Class selection for model execution**
   As a group, the class will choose one scenario from each group to run with the Skive Fjord model. These selected scenarios will be analyzed in detail in the next session.

Seasonal Nutrient Input

Nutrients	Pointsources	Seasonal	Annual - Ø = 0
Oxyg / chl-a	Bottom layer	Summer	↑ / ↓